13EGMO1

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June 9, 2026

Problem

The side BC of the triangle ABC is extended beyond C to D so that $CD = BC$. The side CA is extended beyond A to E so that $AE = 2CA$. Prove that, if $AD = BE$, then the triangle ABC is right-angled.

Solution. Since $EA\colon AC = 2\colon 1$ and $BC = CD$ so A is the centroid of $\triangle EBD$. Let X be the midpoint EB then $2AX = AD = 2BX = 2BX$. So, X is the circumcentre of $\triangle BEA$. Since $E - X - B$, $\angle EAB = 90^\circ$.